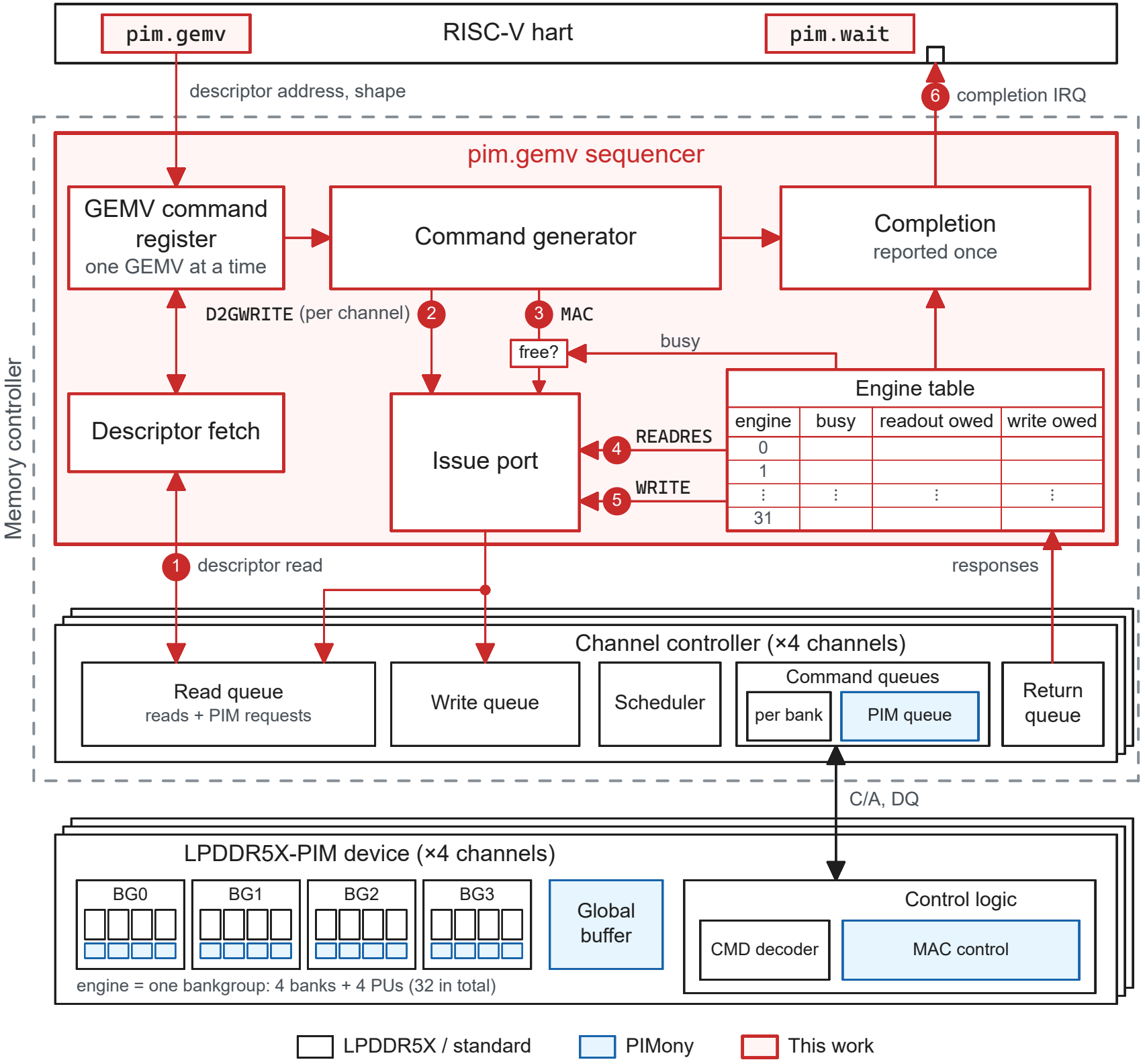

pim.gemv
RISC-V hart
pim.wait
descriptor address, shape
6
completion IRQ
Memory controller
pim.gemv sequencer
GEMV command register
one GEMV at a time
Command generator
Completion
reported once
D2GWRITE (per channel)
2
3
MAC
free?
busy
Descriptor fetch
Issue port
4
READRES
5
WRITE
Engine table
engine
busy
readout owed
write owed
0
1
31
1
descriptor read
responses
Channel controller (×4 channels)
Read queue
reads + PIM requests
Write queue
Scheduler
Command queues
per bank
PIM queue
Return queue
C/A, DQ
LPDDR5X-PIM device (×4 channels)
BG0
BG1
BG2
BG3
Global buffer
Control logic
CMD decoder
MAC control
engine = one bankgroup: 4 banks + 4 PUs (32 in total)
LPDDR5X / standard
PIMony
This work